

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE306	COMPUTER NETWORKS	3	0	0	3	
Course Weightage	ATT: 5 CA: 25 MTT: 20 ETT: 50	Exam Category: 11: Mid Term Exam: All MCQ – End Term Exam: All MCQ				
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT					

Course Outcomes :Through this course students should be able to

CO1 :: Define network topologies, models and functionalities of different layers

CO2 :: Explain data communication, transmission media and modulation at physical layer

CO3 :: Identify error handling and use of multiple access protocols in a network.

CO4 :: Utilize IP addressing and packet forwarding techniques in network setup

CO5 :: Examine different routing algorithms and the performance of unicast routing protocols in network environment

CO6 :: Analyze role and functioning of key application layer protocols and services

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	DATA COMMUNICATIONS AND NETWORKING WITH TCPIP PROTOCOL SUITE	BEHROUZ A. FOROUZAN	Tata McGraw Hill, India

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	COMPUTER NETWORKS	ANDREW S. TANENBAUM	PEARSON

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://nptel.ac.in/courses/106/106/106106091/	Lecture wise content is given on this link
RW-2	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-829-computer-networks-fall-2002/lecture-notes/	MIT university lecture wise content
RW-3	https://www.techtarget.com/searchnetworking/definition/CDN-content-delivery-network	Content Delivery Network
RW-4	https://www.techtarget.com/searchnetworking/definition/SD-WAN-software-defined-WAN	SD-WAN (software-defined WAN)

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Audio Visual Aids (AV)		
Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	https://www.youtube.com/watch?v=VwN91x5i25g	Audio video link for the topics on Computer Networks
AV-2	http://freevideolectures.com/Course/2308/Internet-Technology/6	Video lecture on subnetting
AV-3	https://www.youtube.com/watch?v=sh7fe05mUfA	Introduction to internet, browser, web server, URL, domain name, IP address

Software/Equipments/Databases		
Sr No	(S/E/D) (only if relevant to the course)	Salient Features
SW-1	http://www.packettracernetwork.com/	Packet Tracer
SW-2	http://www.gns3.com/	GNS3

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	7

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	INTRODUCTION(Uses of Computer Networks)	T-1 R-1	RW-1 AV-1	Delivery of lecture zero and describe the basics of network and their applications	Learn about the application of computer network in real life and Learn about application of network in home, business as well as in other areas	Peer to peer discussion, Images in slides, video lecture will be used and Images in slides.	Talk about the various kind of network like mobile network, satellite network, cable TV network and Interaction between any two devices

Week 1	Lecture 1	INTRODUCTION(Uses of Computer Networks)	T-1 R-1	RW-1 AV-1	Delivery of lecture zero and describe the basics of network and their applications	Learn about the application of computer network in real life and Learn about application of network in home, business as well as in other areas	Peer to peer discussion, Images in slides, video lecture will be used and Images in slides.	Talk about the various kind of network like mobile network, satellite network, cable TV network and Interaction between any two devices
	Lecture 2	INTRODUCTION (Introduction to internet, browser, web server, URL, domain name, IP address, packets)	T-1	AV-3	Introduction to internet, browser, web server, URL, domain name, IP address, packets	Make the understanding how a computer network is being used by everyone in their real life.	Peer to peer discussion, animation, videos.	How a user request to server to get access of any webpage and server respond back. Complete process can be discussed..
Week 2	Lecture 3	INTRODUCTION(Network hardware architecture and its topologies and device like HUB, Switch and Routers)	T-1 R-1	RW-1 AV-1	Describes the connecting devices like router, hub and switch.	Learn about the various topology in network as well as types of connections. Learn about the various interconnecting devices	Images in slides, peer to peer discussion images of router, Hub, switch as well as debate on function of these devices	Devices connected in laboratory, internet, cable TV Network Connecting labs with switch and then connect to router for outside world
	Lecture 4	INTRODUCTION(Network software architecture and its layers and protocols)	T-1 R-1	RW-1 AV-1	Describe the protocol hierarchy, design issues,connection oriented, connection less, service primitives	Learn about design issues, protocol hierarchy and various service primitives	Images of protocol Hierarchy, service primitives, peer to peer discussion	Client-server Interaction
Week 3	Lecture 5	NETWORK MODELS (Protocol Layering)	T-1 R-1		Need of layered approach in networking	Learn about the need of layer approach in the networking like the OSI model	Learn about the OSI layer functions	Sending letter through postbox
	Lecture 6	NETWORK MODELS (TCP/IP protocol suite)	T-1 R-1		L-6: Describe the OSI model with functions of each layer L-7: Description of the TCP/IP model and the comparison with OSI model	L-6 Learn about the OSI layer functions L-7 Learn TCP/IP model	L-6 Image of OSI model and its layer, video lecture will be used L-7 Image of TCP/IP model, Video lecture will be used	L-6 sending letter through postbox L-7 Exchange of email, chatting application

Week 3	Lecture 6	NETWORK MODELS(OSI Model)	T-1 R-1		L-6: Describe the OSI model with functions of each layer L-7: Description of the TCP/IP model and the comparison with OSI model	L-6 Learn about the OSI layer functions L-7 Learn TCP/IP model	L-6 Image of OSI model and its layer, video lecture will be used L-7 Image of TCP/IP model, Video lecture will be used	L-6 sending letter through postbox L-7 Exchange of email, chatting application
Week 4	Lecture 7	NETWORK MODELS (TCP/IP protocol suite)	T-1 R-1		L-6: Describe the OSI model with functions of each layer L-7: Description of the TCP/IP model and the comparison with OSI model	L-6 Learn about the OSI layer functions L-7 Learn TCP/IP model	L-6 Image of OSI model and its layer, video lecture will be used L-7 Image of TCP/IP model, Video lecture will be used	L-6 sending letter through postbox L-7 Exchange of email, chatting application
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	Lecture 8	PHYSICAL LAYER: Signal & Media(Basics for Data Communications and Analog and Digital signals)	T-1 R-1	RW-1 AV-1	L-8 :Describe the basics of data and signals L-9: Describes the various data rate methods for the noiseless and noisy channel	L-8 learn the concept of digital and analog signal transmission L-9 Learn about the various methods data rate methods with examples	L-8 Images in slides and peer to peer discussion L-9 Images in slides, video lectures will be used	L-8 downloading speed to download data from internet, Connect the system to exchange data L-9 Wifi, wireless and jamming of signals network

Week 4	Lecture 8	PHYSICAL LAYER: Signal & Media(Data Rate)	T-1 R-1	RW-1 AV-1	L-8 :Describe the basics of data and signals L-9: Describes the various data rate methods for the noiseless and noisy channel	L-8 learn the concept of digital and analog signal transmission L-9 Learn about the various methods data rate methods with examples	L-8 Images in slides and peer to peer discussion L-9 Images in slides, video lectures will be used	L-8 downloading speed to download data from internet, Connect the system to exchange data L-9 Wifi, wireless and jamming of signals network
Week 5	Lecture 9	PHYSICAL LAYER: Signal & Media(Basics for Data Communications and Analog and Digital signals)	T-1 R-1	RW-1 AV-1	L-8 :Describe the basics of data and signals L-9: Describes the various data rate methods for the noiseless and noisy channel	L-8 learn the concept of digital and analog signal transmission L-9 Learn about the various methods data rate methods with examples	L-8 Images in slides and peer to peer discussion L-9 Images in slides, video lectures will be used	L-8 downloading speed to download data from internet, Connect the system to exchange data L-9 Wifi, wireless and jamming of signals network
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	Lecture 10	PHYSICAL LAYER: Signal & Media(Transmission Impairments and Performance)	T-1 R-1	RW-1 AV-1	Describes the Transmission impairment issues and guided and unguided media	Learn about the Transmission impairment issues during transmission as well as guided and unguided media	Images in slides, peer to peer discussion	Disturbance while listening radio

Week 5	Lecture 10	PHYSICAL LAYER: Signal & Media(Transmission media like Guided and Unguided media)	T-1 R-1	RW-1 AV-1	Describes the Transmission impairment issues and guided and unguided media. Also cabling standards need to be discuss.	Learn about the Transmission impairment issues during transmission as well as guided and unguided media	Images in slides, peer to peer discussion	Disturbance while listening radio
		PHYSICAL LAYER: Signal & Media(Cabling standards)	T-1 R-1	RW-1 AV-1	Describes the Transmission impairment issues and guided and unguided media. Also cabling standards need to be discuss.	Learn about the Transmission impairment issues during transmission as well as guided and unguided media	Images in slides, peer to peer discussion	Disturbance while listening radio
Week 6	Lecture 11	PHYSICAL LAYER: Modulation & Multiplexing (Digital to Digital Conversion)	T-1 R-1	RW-1	L-11: Describes the Line coding technique (NRZ, NRZ-L, NRZ-I, RZ, manchester and differential manchester) L- 12: PCM and analog to analog conversion Technique L-13:Describes the modulation technique (ASK,PSK,FSK)	L-11 Learn about line coding techniques L-12 Learn about the PCM and AM, FM and PM techniques L-13 Learn about the digital to analog conversion methods	L-11 images in slides and peer to peer discussion L-12 images in slides and brain storming L-13 video lecture and images in slides will be used	L-11 Binary data encoded into signals L-12 Modem, Telephone network L-13 Cable TV network
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	Lecture 14	PHYSICAL LAYER: Modulation & Multiplexing (Multiplexing)	T-1 R-1		Describes the multiplexing (FDM,TDM and its types)	Learn about the multiplexing and importance in data communication	video lecture and images in slides will be used	PSTN
Week 8	Lecture 15				Test 1			
	Lecture 16	DATA LINK LAYER(Data link Layer design issues)	T-1 R-1	RW-2	Describes bout the working of switch along with the design related issues in data link layer	Learn about the design issues of data link layer	Images in slides, peer to peer discussion	sender speed need to synchronize with receiver, uplifting and down lifting water from tank
		DATA LINK LAYER(Error Detection and Correction- Hamming code, CRC, Parity, Checksum)	T-1 R-1	RW-2	Describes about the various error deduction and correction methods	Learn about error deduction and correction methods	Images in slides, peer to peer discussion	Hamming code
		DATA LINK LAYER (Switch working)	T-1 R-1	RW-2	Describes bout the working of switch along with the design related issues in data link layer	Learn about the design issues of data link layer	Images in slides, peer to peer discussion	sender speed need to synchronize with receiver, uplifting and down lifting water from tank

Week 9	Lecture 17	DATA LINK LAYER (Elementary Datalink Protocols)	T-1 R-1	RW-1 AV-1	Describes the Simplex Stop and wait for noisy and noiseless channel, describes the sliding window protocol like go back n and selective repeat protocol	Learn about the flow control and error control protocol like stop and wait, stop and wait ARQ and error control protocol like go back n and selective repeat protocol	Images of stop and wait protocol for noisy and noiseless channel, Peer to peer discussion and images of Go back n and selective repeat protocol in slides, Peer to peer discussion	a) one person talking to other over mobile phone, while other is listening so once one person completed after that listener will respond or sender will wait for respond of other person b) person can send the bulk of data at a time with out waiting respond from other person
	Lecture 18	DATA LINK LAYER (Elementary Datalink Protocols)	T-1 R-1	RW-1 AV-1	Describes the Simplex Stop and wait for noisy and noiseless channel, describes the sliding window protocol like go back n and selective repeat protocol	Learn about the flow control and error control protocol like stop and wait, stop and wait ARQ and error control protocol like go back n and selective repeat protocol	Images of stop and wait protocol for noisy and noiseless channel, Peer to peer discussion and images of Go back n and selective repeat protocol in slides, Peer to peer discussion	a) one person talking to other over mobile phone, while other is listening so once one person completed after that listener will respond or sender will wait for respond of other person b) person can send the bulk of data at a time with out waiting respond from other person
Week 10	Lecture 19	MAC SUBLAYER(Multiple Access Protocols- ALOHA, CSMA and CSMA/CD)	T-1 R-1	RW-1	Describe the random access protocol like the CSMA/CD and CSMA/CA along with the controlled access methods like the reservation and polling	Learn the CSMA and CSMA/CD protocol	Brain storming and Images in slides	Crossing the road with out looking traffic

Week 10	Lecture 19	MAC SUBLAYER(Random Access)	T-1 R-1	RW-1	Describe the random access protocol like the CSMA/CD and CSMA/CA along with the controlled access methods like the reservation and polling	Learn the CSMA and CSMA/CD protocol	Brain storming and Images in slides	Crossing the road with out looking traffic
		MAC SUBLAYER (Controlled access)	T-1 R-1	RW-1	Describe the random access protocol like the CSMA/CD and CSMA/CA along with the controlled access methods like the reservation and polling	Learn the CSMA and CSMA/CD protocol	Brain storming and Images in slides	Crossing the road with out looking traffic
		MAC SUBLAYER(Ethernet protocol)	T-1 R-1	RW-1	Describe the random access protocol like the CSMA/CD and CSMA/CA along with the controlled access methods like the reservation and polling	Learn the CSMA and CSMA/CD protocol	Brain storming and Images in slides	Crossing the road with out looking traffic
		SPILL OVER						
Week 10	Lecture 20				Spill Over			
Week 11	Lecture 21				Spill Over			
		MID-TERM						
Week 11	Lecture 22	NETWORK LAYER: IP Addressing(Network layer design issue)	T-1 R-1	RW-1 AV-1	Describes the Network layer design issues and network layer services	Learn about the network layer issues and services	Images in slides and video lecture will be used	
		NETWORK LAYER: IP Addressing(Network layer services)	T-1 R-1	RW-1 AV-1	Describes the Network layer design issues and network layer services	Learn about the network layer issues and services	Images in slides and video lecture will be used	
Week 12	Lecture 23	NETWORK LAYER: IP Addressing(IP Addressing Both Classfull and Classless)	T-1 R-1	RW-1 SW-1 AV-1	Describes the Classfull and class less IP addressing along with network layer performance metrics like the delay, throughput and packet loss	Learn about the basics of classfull and classless IP addressing. Also student will know about the metrics for the network design	Images in slides and use of white board	

Week 12	Lecture 23	NETWORK LAYER: IP Addressing(Network layer performance)	T-1 R-1	RW-1 SW-1 AV-1	Describes the Classfull and class less IP addressing along with network layer performance metrics like the delay, throughput and packet loss	Learn about the basics of classfull and classless IP addressing. Also student will know about the metrics for the network design	Images in slides and use of white board	
		NETWORK LAYER: IP Addressing(Forwarding of IP packets)	T-1 R-1	RW-1 SW-1 AV-1	Describes the Classfull and class less IP addressing along with network layer performance metrics like the delay, throughput and packet loss	Learn about the basics of classfull and classless IP addressing. Also student will know about the metrics for the network design	Images in slides and use of white board	
	Lecture 24	NETWORK LAYER: IP Addressing(Subnetting and Supernetting)	T-1 R-1	RW-1 SW-2 AV-1 AV-2	L- 24: Describes the idea of subnetting as well as solve example based on subnetting. L-25: Also the basic description of IP header with IPv6 addressing is described	L-24 Learn the concept of subnetting and solve problem based on it L25- IP header fields and the working of IPv6	Use of white board to solve problems and ask students to solve problem	
		NETWORK LAYER: IP Addressing(Subnetting examples)	T-1 R-1	RW-1 SW-2 AV-1 AV-2	L- 24: Describes the idea of subnetting as well as solve example based on subnetting. L-25: Also the basic description of IP header with IPv6 addressing is described	L-24 Learn the concept of subnetting and solve problem based on it L25- IP header fields and the working of IPv6	Use of white board to solve problems and ask students to solve problem	
		NETWORK LAYER: IP Addressing(IP Header)	T-1 R-1	RW-1 SW-2 AV-1 AV-2	L- 24: Describes the idea of subnetting as well as solve example based on subnetting. L-25: Also the basic description of IP header with IPv6 addressing is described	L-24 Learn the concept of subnetting and solve problem based on it L25- IP header fields and the working of IPv6	Use of white board to solve problems and ask students to solve problem	

Week 12	Lecture 24	NETWORK LAYER: IP Addressing(IPv6 addressing)	T-1 R-1	RW-1 SW-2 AV-1 AV-2	L- 24: Describes the idea of subnetting as well as solve example based on subnetting. L-25: Also the basic description of IP header with IPv6 addressing is described	L-24 Learn the concept of subnetting and solve problem based on it L25- IP header fields and the working of IPv6	Use of white board to solve problems and ask students to solve problem	
Week 13	Lecture 25	NETWORK LAYER: IP Addressing(Subnetting and Supernetting)	T-1 R-1	RW-1 SW-2 AV-1 AV-2	L- 24: Describes the idea of subnetting as well as solve example based on subnetting. L-25: Also the basic description of IP header with IPv6 addressing is described	L-24 Learn the concept of subnetting and solve problem based on it L25- IP header fields and the working of IPv6	Use of white board to solve problems and ask students to solve problem	
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		NETWORK LAYER: IP Addressing(IPv6 addressing)	T-1 R-1	RW-1 SW-2 AV-1 AV-2	L- 24: Describes the idea of subnetting as well as solve example based on subnetting. L-25: Also the basic description of IP header with IPv6 addressing is described	L-24 Learn the concept of subnetting and solve problem based on it L25- IP header fields and the working of IPv6	Use of white board to solve problems and ask students to solve problem	

Week 13	Lecture 26	NETWORK LAYER: IP Addressing(IPv6 addressing)	T-1 R-1	RW-1 SW-2 AV-1 AV-2	L- 24: Describes the idea of subnetting as well as solve example based on subnetting. L-25: Also the basic description of IP header with IPv6 addressing is described	L-24 Learn the concept of subnetting and solve problem based on it L25- IP header fields and the working of IPv6	Use of white board to solve problems and ask students to solve problem	
Week 14	Lecture 27	NETWORK LAYER: Routing(Routing algorithms)	T-1 R-1	RW-1	L-27: Describes the basics of routing and types L-28: Description of unicast routing algo like the RIP, OSPF, BGP in-terms of only metrics and advantages	Students will aware about the routing basics and the metrics associated with them	PPT and the animations for demonstrating the routing	
		NETWORK LAYER: Routing(Unicast routing protocols)	T-1 R-1	RW-1	L-27: Describes the basics of routing and types L-28: Description of unicast routing algo like the RIP, OSPF, BGP in-terms of only metrics and advantages	Students will aware about the routing basics and the metrics associated with them	PPT and the animations for demonstrating the routing	
	Lecture 28	NETWORK LAYER: Routing(Routing algorithms)	T-1 R-1	RW-1	L-27: Describes the basics of routing and types L-28: Description of unicast routing algo like the RIP, OSPF, BGP in-terms of only metrics and advantages	Students will aware about the routing basics and the metrics associated with them	PPT and the animations for demonstrating the routing	
		NETWORK LAYER: Routing(Unicast routing protocols)	T-1 R-1	RW-1	L-27: Describes the basics of routing and types L-28: Description of unicast routing algo like the RIP, OSPF, BGP in-terms of only metrics and advantages	Students will aware about the routing basics and the metrics associated with them	PPT and the animations for demonstrating the routing	
Week 15	Lecture 29				Test 2			

Week 15	Lecture 30	NETWORK LAYER: Routing(Routing Algorithm- Shortest path algorithm)	T-1 R-1	RW-1	L-30: Describes the optimality principle, shortest path algorithm, flooding L-30: Describes the distance vector, link state algorithm	L-30 Learn about the types of routing algorithm and flooding L-30 Learn about the distance vector, link state algorithm	L-30 Images in slides, peer to peer discussion L-30 Images in slides as well as use white board to form routing table	
		NETWORK LAYER: Routing(Distance vector Routing)	T-1 R-1	RW-1	L-30: Describes the optimality principle, shortest path algorithm, flooding L-30: Describes the distance vector, link state algorithm	L-30 Learn about the types of routing algorithm and flooding L-30 Learn about the distance vector, link state algorithm	L-30 Images in slides, peer to peer discussion L-30 Images in slides as well as use white board to form routing table	
		NETWORK LAYER: Routing(Link State routing)	T-1 R-1	RW-1	L-29: Describes the optimality principle, shortest path algorithm, flooding L-30: Describes the distance vector, link state algorithm	L-29 Learn about the types of routing algorithm and flooding L-30 Learn about the distance vector, link state algorithm	L-29 Images in slides, peer to peer discussion L-30 Images in slides as well as use white board to form routing table	
Week 16	Lecture 31	NETWORK LAYER: Congestion Control (Congestion Control Algorithms)	T-1 R-1	RW-1 RW-2	L-31: Describe the congestion,congestion control, approaches to congestion control (Network provisioning,traffic aware routing,admission control) and Describe the approaches to congestion control algorithm (Traffic throttling, choke packet, hop by hop back pressure) L-32 :Describes the quality of service, traffic shaping, leaky bucket, token bucket, packet scheduling (FIFO, round robin, weighted fair queueing)	L-31Learn about the congestion and methods to control the congestion in network and also Learn about the methods to control the congestion in network L-32 Learn about the quality of service and methods to maintain quality of service in network	L-31Images in slides and peer to peer discussion L-32 congestion control methods Images, peer to peer discussion, brain storming	

Week 16	Lecture 32	NETWORK LAYER: Congestion Control (Congestion Control Algorithms)	T-1 R-1	RW-1 RW-2	L-31: Describe the congestion, congestion control, approaches to congestion control (Network provisioning, traffic aware routing, admission control) and Describe the approaches to congestion control algorithm (Traffic throttling, choke packet, hop by hop back pressure) L-32 :Describes the quality of service, traffic shaping, leaky bucket, token bucket, packet scheduling (FIFO, round robin, weighted fair queueing)	L-31 Learn about the congestion and methods to control the congestion in network and also Learn about the methods to control the congestion in network L-32 Learn about the quality of service and methods to maintain quality of service in network	L-31 Images in slides and peer to peer discussion L-32 congestion control methods Images, peer to peer discussion, brain storming	
Week 17	Lecture 33	TRANSPORT LAYER (Transport Layer Services)	T-1 R-1	RW-1 RW-2 AV-1	L-33 : Abstract of the transport layer services and classification of Port number, connection oriented and connection-less communication (overview only) L-34:Describes TCP and UDP header format, three way handshaking (overview only)	L33 - Learn about the services provided by transport layer, connection oriented and connection-less communication L34 - Learn about the Importance of TCP and UDP header fields and handshaking operation	L-33 Image of TCP header, peer to peer discussion L-34 Image of TCP and UDP header, Images of three way Handshaking operation, peer to peer discussion	
		TRANSPORT LAYER (TCP- Header format and handshaking operation)	T-1 R-1	RW-1 RW-2 AV-1	L-33 : Abstract of the transport layer services and classification of Port number, connection oriented and connection-less communication (overview only) L-34:Describes TCP and UDP header format, three way handshaking (overview only) and UDP header	L33 - Learn about the services provided by transport layer, connection oriented and connection-less communication L34 - Learn about the Importance of TCP and UDP header fields and handshaking operation	L-33 Image of TCP header, peer to peer discussion L-34 Image of TCP and UDP header, Images of three way Handshaking operation, peer to peer discussion	

Week 17	Lecture 33	TRANSPORT LAYER (UDP- Header format)	T-1 R-1	RW-1 RW-2 AV-1	L-33 : Abstract of the transport layer services and classification of Port number, connection oriented and connection-less communication (overview only) L-34:Describes TCP and UDP header format, three way handshaking (overview only) and UDP header	L33 - Learn about the services provided by transport layer, connection oriented and connection-less communication L34 - Learn about the Importance of TCP and UDP header fields and handshaking operation	L-33 Image of TCP header, peer to peer discussion L-34 Image of TCP and UDP header, Images of three way Handshaking operation, peer to peer discussion	
	Lecture 34	TRANSPORT LAYER (Transport Layer Services)	T-1 R-1	RW-1 RW-2 AV-1	L-33 : Abstract of the transport layer services and classification of Port number, connection oriented and connection-less communication (overview only) L-34:Describes TCP and UDP header format, three way handshaking (overview only)	L33 - Learn about the services provided by transport layer, connection oriented and connection-less communication L34 - Learn about the Importance of TCP and UDP header fields and handshaking operation	L-33 Image of TCP header, peer to peer discussion L-34 Image of TCP and UDP header, Images of three way Handshaking operation, peer to peer discussion	
		TRANSPORT LAYER (TCP- Header format and handshaking operation)	T-1 R-1	RW-1 RW-2 AV-1	L-33 : Abstract of the transport layer services and classification of Port number, connection oriented and connection-less communication (overview only) L-34:Describes TCP and UDP header format, three way handshaking (overview only) and UDP header	L33 - Learn about the services provided by transport layer, connection oriented and connection-less communication L34 - Learn about the Importance of TCP and UDP header fields and handshaking operation	L-33 Image of TCP header, peer to peer discussion L-34 Image of TCP and UDP header, Images of three way Handshaking operation, peer to peer discussion	

Week 17	Lecture 34	TRANSPORT LAYER (UDP- Header format)	T-1 R-1	RW-1 RW-2 AV-1	L-33 : Abstract of the transport layer services and classification of Port number, connection oriented and connection-less communication (overview only) L-34:Describes TCP and UDP header format, three way handshaking (overview only) and UDP header	L33 - Learn about the services provided by transport layer, connection oriented and connection-less communication L34 - Learn about the Importance of TCP and UDP header fields and handshaking operation	L-33 Image of TCP header, peer to peer discussion L-34 Image of TCP and UDP header, Images of three way Handshaking operation, peer to peer discussion	
Week 18	Lecture 35	RECENT TRENDS IN COMPUTER NETWORKS (Software Defined Wide Area Network (SDWAN))	T-1	RW-4	Overview of SD-WAN architecture, benefits, and its role in modern network management, Introduction to CDN, its components, and its importance in delivering content efficiently over the internet	Understand the functionality of SD-WAN, Learn how CDNs reduce latency, improve content delivery speed, and enhance user experiences globally	Demonstration and Peer Discussion	Case study of a retail chain using SD-WAN, Example of a video streaming platform using CDN
		RECENT TRENDS IN COMPUTER NETWORKS (Content Delivery Network (CDN))	T-1	RW-3	Overview of SD-WAN architecture, benefits, and its role in modern network management, Introduction to CDN, its components, and its importance in delivering content efficiently over the internet	Understand the functionality of SD-WAN, Learn how CDNs reduce latency, improve content delivery speed, and enhance user experiences globally	Demonstration and Peer Discussion	Example of a video streaming platform using CDN
	Lecture 36	APPLICATION LAYER(E Mail)	T-1 R-1	RW-1 AV-1	Describes the email architecture	Learn about the email architecture	Image of Email Architecture, peer to peer discussion	
Week 19	Lecture 37	APPLICATION LAYER(E Mail)	T-1 R-1	RW-1 AV-1	Describes the email architecture	Learn about the email architecture	Image of Email Architecture, peer to peer discussion	
	Lecture 38	APPLICATION LAYER (Domain Name System)	T-1 R-1	RW-1 AV-1	Domain name system Hierarchy (Overview only)and FTP	Learn about the Domain name system hierarchy and FTP	Image of DNS server and query resolving operation	
Week 20	Lecture 39	APPLICATION LAYER (Domain Name System)	T-1 R-1	RW-1 AV-1	Domain name system Hierarchy (Overview only)and FTP	Learn about the Domain name system hierarchy and FTP	Image of DNS server and query resolving operation	

Week 20	Lecture 40	APPLICATION LAYER (FTP)	T-1 R-1	RW-1 AV-1	Domain name system Hierarchy (Overview only)and FTP	Learn about the Domain name system hierarchy and FTP	Image of DNS server and query resolving operation	
		SPILL OVER						
Week 21	Lecture 41				Spill Over			
	Lecture 42				Spill Over			
Week 22	Lecture 43				Spill Over			
	Lecture 44				Spill Over			
Week 23	Lecture 45				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Test 2	50	
Test 1	50	
Recoup CA		

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Test 2	Assess knowledge of IP addressing, subnetting, routing algorithms, and congestion control in network layer operations.	Subjective type evaluation having 30 Marks: Two questions of 10 Marks each and Two Questions of 5 Marks each.	Individual	Offline	30	11 / 13

Test 1	Assess understanding of network components, protocols, physical layer signals, media, and transmission techniques	30 MCQs having 1 mark each	Individual	Online	30	3 / 5
Recoup CA	subjective test based on the syllabus	Deactivated 10June 25Additional chance of CA	Individual	Online	30	8 / 10